

Predictive and Explainable Forecasting of Financial and ESG Performance in Private Companies: Evidence from Campania (2015–2024)

Cristina Arino, Alessandra D’Arienzo, Lucio Di Gioia
University of Napoli “Federico II” (Italy)
{cr.arino, ale.darienzo, lucio.digioia}@studenti.unina.it

Abstract—Anticipating the economic and sustainability trajectories of private companies is increasingly important for investors, financial institutions, and policymakers, yet private company data are often heterogeneous and affected by missingness, outliers, and inconsistent reporting. This work addresses this gap by proposing a machine learning framework that jointly forecasts key financial indicators and ESG performance while providing interpretable, cluster-level insights. Our pipeline combines robust preprocessing, feature scaling, and forecasting. Using a company level dataset covering companies located in the Campania region over the period 2015–2024, we benchmark multiple regression and time series models, including Linear Regression, Random Forest, XGBoost, Long Short-Term Memory (LSTM) networks, and Prophet, to forecast key financial indicators and ESG scores. Empirically, Random Forest achieves the lowest error ($MSE = 0.00285$), closely followed by LSTM ($MSE = 0.00287$), while Prophet performs worst ($MSE = 0.00392$). Model performance is stable across macro-areas and macro-sectors, whereas error varies by firm size (micro firms: $MSE = 0.00076$; small: 0.00219 ; medium-large: 0.00505). Building on the selected Random Forest, we generate 2024 predictions for Sales Revenues, EBITDA, Net Income, and ESG, and propose a Gartner-style positioning diagram to identify companies with jointly high predicted economic and sustainability performance, with notable concentration in the Industry and Construction macro-sector. The proposed framework provides a scalable and interpretable tool for analyzing private companies and supports data-driven decision making in heterogeneous economic environments.

Index Terms—Machine learning, ESG prediction, Private companies, Financial forecasting, Sustainable investment.

I. INTRODUCTION

In recent years, the ability to anticipate the economic and financial trajectories of private companies has become increasingly relevant to a wide range of stakeholders, including entrepreneurs, investors, lenders, supply-chain partners, and policymakers, also due to increased uncertainty and stronger sustainability-related requirements [1, 2]. Persistent macroeconomic uncertainty, tighter financial conditions, and the acceleration of digital and sustainability transitions have increased the value of analytical tools that support strategic planning, risk management, and resource allocation. At the same time, the growing availability of firm-level data enables the development of predictive systems that can complement traditional performance monitoring, especially for the heterogeneous and dynamic population of privately held companies. In this context, Machine Learning (ML) and Deep Learning

(DL) techniques have gained traction due to their ability to model interactions and non-linearities with fewer functional-form assumptions.

Private-owned companies introduce additional empirical and methodological challenges. Compared to publicly listed companies, they are typically characterized by greater heterogeneity in scale, governance structures, and strategic objectives, as well as less standardized disclosure and a potentially higher incidence of missingness, outliers, and reporting discontinuities in available data. These aspects make data cleaning, temporal consistency checks, and validation design critical components of any predictive pipeline. Moreover, while ESG indicators are becoming more prevalent also among private firms, driven by regulatory spillovers, supply-chain requirements, and financing conditions, their relationship with economic-financial performance remains context-dependent and potentially non-linear, varying across sectors, territories, and company sizes. Consequently, there is a need for research that combines robust preprocessing, high-performing predictive models, and interpretability tools, explicitly accounting for heterogeneity in private companies populations.

This paper presents an end-to-end framework for the predictive and explainable analysis of economic performance in private companies, developed within a data-driven research challenge focused on companies located in the Italian region of Campania. Using a firm-level dataset spanning 2015–2024 and integrating both financial and ESG indicators, we propose a workflow that includes: (i) data cleaning and missing-value treatment based on state-of-the-art methodologies; (ii) benchmarking of multiple regression algorithms to identify the most accurate predictive model; (iii) forward-looking forecasting of performance metrics; and (iv) interpretability and correlation analysis to uncover statistically significant relationships between financial outcomes and ESG dimensions. A distinctive element of our approach is the systematic organization of results into meaningful clusters including company size, macro-sector, and geographic aggregation thereby enabling stakeholder-oriented interpretation, providing a useful tool to identify possible trends and responsible and sustainable investment opportunities.

The remainder of the paper is organized as follows. Sec. II reviews state-of-the-art methodologies that leverage ML and DL to predict firms’ economic indicators and to assess

their relationships with ESG dimensions. Sec. III presents the methodology adopted in this study. Sec. IV introduces the dataset and details the preprocessing pipeline, input formatting, and evaluation criteria. Sec. V reports the empirical results covering predictive performance, interpretability, and correlation analysis with a particular emphasis on cluster-level insights and managerial implications. Finally, Sec. VI concludes the paper and outlines directions for future research.

II. RELATED WORK

Recent literature on data-driven company performance analysis has evolved along two closely related research streams. The first focuses on predicting economic and financial indicators through ML and DL techniques, the second investigates the role of ESG dimensions either by studying their relationship with performance outcomes or by predicting ESG scores from company level structural information and other covariates. The present section summarizes these directions and clarifies how our work positions itself relative to existing contributions.

A wide body of work shows that ML models can effectively forecast companies' financial indicators by exploiting patterns embedded in historical accounting information. Existing studies commonly benchmark multiple supervised algorithms, including linear regression, decision trees, random forests, support vector regression, and gradient boosting methods, with the objective of improving predictive accuracy compared to traditional parametric specifications. For instance, Guarino et al. [3] employ a sliding window approach and benchmark several regression algorithms (Linear Regression, Decision Tree, Random Forest (RF), and XGBoost), highlighting that performance can differ between groups defined by firm characteristics such as size, sector, and geography.

As highlighted in the systematic review by Sezer et al. [4], DL models and in particular Recurrent Neural Networks based on Long Short-Term Memory (LSTM) units have emerged as a state-of-the-art approach for financial time series forecasting. Following this line of research, our study complements tree-based and linear models with an LSTM-based approach to explicitly model the temporal structure of firms' financial indicators.

A parallel stream of research addresses ESG from two complementary angles. First, ESG as a driver of economic performance. For example, panel-data evidence on Malaysian and Singaporean companies (2010–2014) [5] reports significant positive correlations between ESG dimensions and a “construct of “economic performance””, with heterogeneous effects between countries and pillars. At the macro level, Diaye et al. [6] investigate ESG performances as a determinant of economic growth, with panel cointegration evidence on OECD countries suggesting that positive effects can emerge in the long run even when short-run relationships are weak or heterogeneous. Second, a number of studies have concentrated in ESG prediction via ML. For example, D'Amato et al. [7] apply RF regression to explain Refinitiv ESG scores for STOXX Europe 600 constituents (2009–2019), showing

that balance-sheet items provide substantial explanatory power and motivating the use of ML. More recently, ESG rating prediction has been formulated as a multiclass classification problem at global scale: a study using 6,166 firms across 73 countries (2005–2019) compares six ML algorithms and finds that a Random Forest classifier achieves the highest accuracy, while variable-importance analysis highlights the role of lagged ESG scores and firm characteristics such as size and leverage [8]. These findings collectively indicate both the feasibility of data-driven ESG modeling and the central role of tree-based ensembles in ESG prediction tasks.

Positioning. Building on these research directions, our paper aims to identify the most accurate approach for predicting both economic and ESG performance in private companies within a localized economic context. In addition to related literature, we provide deeper insight into the ESG–financial relationship through correlation analysis and cluster-based evidence.

III. METHODOLOGY

In this section, we present the methodology adopted in our study. Specifically, Section III-A defines the prediction tasks and the models considered, while Section III-B describes the GenAI-based approach used to address missing values.

A. Prediction Methodology

Problem Statement. We aim to predict P financial and ESG indicators for each company using the historical observations of the same indicators. Let $X = \{x_n\}_{n=1}^N$ denote a multivariate time series, where $x_n = \{x_{n,p}\}_{p=1}^P$ represents the value of the p -th financial indicator observed at time instant n . Our objective is to predict the value of the p -th indicator at time $n + 1$, denoted by $\hat{x}_{n+1,p}$.

Models Considered. Here, we introduce the ML and DL models adopted for the prediction task:

- **Linear Regression (LR):** a baseline model that estimates the target indicator as a linear combination of the input features. It is easy to train and interpret, but it may struggle to capture non-linear relationships and complex interactions;
- **Random Forest Regressor (RFR):** an ensemble method that combines multiple decision trees trained on bootstrapped samples and random subsets of features. Since single decision trees are typically unstable and sensitive to data perturbations, RFR reduces variance by averaging the predictions of the individual trees;
- **XGBoost (XGB):** another ensemble method based on the concept of boosting rather than on bagging. It builds trees sequentially, each correcting the errors of the previous ones. Ensemble methods are best known for their high performances, even though the main disadvantage is that they are slower to train.
- **Prophet Algorithm (PA):** a time-series forecasting model that represents the signal as an additive combination of trend, seasonality, and, when available, holiday effects. It is designed to be robust to missing data and outliers

and is particularly suitable when the series exhibits clear trend and seasonal patterns [9].

- Long Short-Term Memory (LSTM): a recurrent neural network architecture tailored for sequential data, capable of learning long-range temporal dependencies through gated memory cells. It can model complex non-linear dynamics in time series, but typically requires more data and careful hyperparameter tuning to generalize well.

B. Handling of Missing Values via AI Application

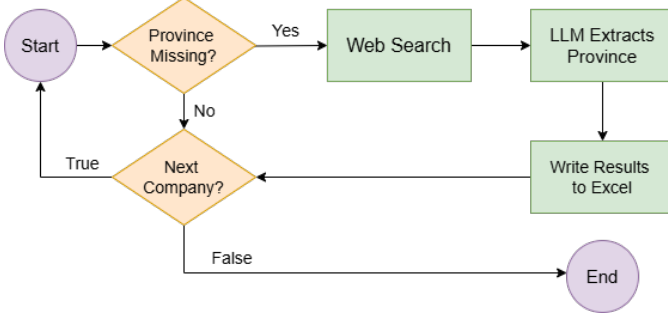


Fig. 1: Schematic overview of the GenAI-based application.

To address missing values, we developed an AI application that integrates a Large Language Model (LLM) with a web-search tool. For each query, the application invokes the search tool to retrieve relevant information and then leverages the LLM to synthesize and return the desired output.

To mitigate hallucinations and improve inference reliability, we adopt an iterative verification and aggregation mechanism based on majority voting. The underlying rationale is to prevent a single generation, that might be potentially affected by noisy search results or residual stochasticity, from directly determining the imputed value. Instead, the system favors a final output supported by coherence checks. A synthetic schematic overview of the AI application workflow is reported in Fig. 1.

IV. EXPERIMENTAL SETUP

Here, we describe the experimental setup adopted. We first introduce the dataset used in this study and then detail the preprocessing pipeline implemented. Next, we describe how the cleaned dataset is transformed into model-ready inputs, specifying the adopted input formatting strategy for each family of predictors as well as the train/test split design used for prediction. Finally, we present the performance evaluation criteria employed to benchmark models consistently, enabling a fair comparison across alternative ML and DL approaches.

A. Dataset & Pre-Processing Pipeline.

Dataset Description. Our work starts from a dataset including 12,422 samples of private companies located in the Campania region each described by 117 distinct features provided by Iniziativa Cube S.r.l., an Italian consulting firm specialized in finance. For each company, we were provided with basic information such as Company Name, ATECO Code, ATECO

Description, Operational Headquarters Address (Municipality and Province), Fiscal Code, Trucost Sector, Trucost Sector Description and annual Employers Count. The core of our predictive task focuses on a total of *ten indicators* between financial spanning 2015-2024 and ESG-related variables (2019-2021), presented in detail in the following:

- Sales Revenues (SR): total income generated from the company’s core sales over the fiscal year;
- Production Costs (PC): operating costs incurred to produce goods/services;
- Total Assets (TA): total value of the resources controlled by the firm at year end;
- Shareholders’ Equity (SE): owners’ residual interest in the company, equal to assets minus liabilities;
- Net financial position (NFP): net debt measure capturing financial liabilities minus cash and other liquid financial assets;
- PFN/EBITDA: leverage indicator expressing net financial position relative to EBITDA, commonly used to assess debt sustainability;
- Earnings before Interest, Taxes, Depreciation, and Amortization (EBITDA): a proxy of operating profitability and cash generation from core operations;
- Net Profit (NP): bottom-line profit (or loss) after operating and non-operating items, taxes, and financial costs;
- Return on Investment (ROI): profitability ratio measuring operating return relative to invested capital;
- ESG score: composite indicator summarizing the firm’s environmental, social, and governance performance. We are also provided with *six explanatory factors* (Greenhouse gases, Water, Waste, Air pollutants, Land and water pollutants, Natural resources used) whose sum is equal to the ESG score;

After preprocessing, the dataset was refined to include 7,660 samples, each described by 97 distinct variables. The reduction in the number of features is due to the removal of the columns associated with the ROI and PFN/EBITDA indicators. These variables were highly correlated with other predictors already included in the dataset, provided limited additional information for prediction, and exhibited a high proportion of missing values.

Management of Data Quality Issues. In this paragraph, we outline the preprocessing procedure adopted to address common data-quality issues in company records, such as missing values, outliers, and inconsistencies, and to ensure comparability across firms and years.

First, we inspected the dataset for potentially incoherent values, including zeros and negative entries. For SR, PC and TA values less or equal to zero are not economically feasible and their presence in the original dataset might reflect recording errors or placeholders. Therefore, we treated these observations by setting the corresponding cells to Not a Number (NaN).

Second, we addressed outliers, which can substantially distort the analysis given the strong heterogeneity across companies. For the economic indicators, we adopted a percentile-

based filtering strategy by removing observations falling in the top and bottom 1% of the distribution for each financial indicator and each year. This procedure led to the removal of 3,043 entries from the dataset. In contrast, ESG outliers were handled via 95% winsorization: ESG scores above the 95th percentile were replaced with the corresponding 95th-percentile value.

Finally, we addressed missing values, which are critical for the proper training and evaluation of most predictive algorithms. Before filtering out firms with extensive missingness, we assessed whether part of the missing information could be recovered. In particular, we observed that 1,304 records lacked the *operational headquarters province* field. To fill this gap, we developed an AI application integrating an LLM with a web-search tool. Specifically, we used the *Llama-3.1-8B-Instant* model (via Groq) and *DuckDuckGo* as the search engine. For each incomplete record, the application queries the web and leverages the LLM to aggregate and validate the retrieved evidence, returning an imputed province value.

After that, we removed companies exhibiting more than 40% missing values across all features, reducing the sample from 9,152 to 8,040 firms. For the remaining companies, we imputed residual missing values through linear interpolation along the temporal dimension: the last year was imputed using the previous year’s value, the first year using the second year’s value, and intermediate gaps by averaging adjacent years. At the end of this preprocessing pipeline, 385 companies still exhibited missing values for the Net Financial Position (NFP) indicator across all years; for simplicity, we removed these observations. The final dataset therefore includes 7,660 companies.

Management of Categorical Variables. In order to guarantee a focused and comparable interpretation of prediction results, we organized companies into three clusters: (a) Company Size, (b) Geographic Macro-Area and (c) Macro-Sector. For "Company Size", we defined three groups: 0-9 (micro), 10-49 (small), 50+ (medium-large), with small companies representing the largest share of the sample. For the "Geographic Macro-Area", we assigned companies to the five provinces of the Campania region using the AI application described in Section III-B. For the "Macro-Sector", companies were mapped from their ATECO classification into five broad categories: Industry and Construction; Trade and Consumer Services; Networks, Logistics and Infrastructure; Knowledge Economy and Advanced Services; and Finance, Real Estate and Collective Services. The first two macro-sectors are the most represented in the dataset.

Input Data Formatting. To ensure a consistent scale across variables and improve training stability, we normalized all numeric feature before feeding them into the predictive models. We adopted a two step scaling procedure. First, we applied the *Robust Scaler*, which rescales each feature by subtracting its median and dividing by the interquartile range (IQR). This transformation is less sensitive to extreme values than standard normalization and is therefore better suited to the heterogeneous data at our disposal, where residual outliers persist even

after filtering. Second, we applied *Min-Max scaling* to map the robust-scaled features into a fixed range, typically $[0, 1]$. This step enforces comparable numerical bounds across features, which is particularly beneficial for gradient-based learners and neural networks, and facilitates stable optimization during training.

B. Performance Evaluation Criteria.

We split the dataset into training and test sets using a 75/25 ratio and stratified the split to preserve a balanced distribution of company profiles with respect to company size, which is the categorical variable capturing the largest heterogeneity across firms. The predictive models introduced in Section III-A were evaluated using the Mean Squared Error (MSE) metric. For each model, the main hyperparameters were fine-tuned over predefined ranges in order to achieve the best possible performance. All baseline ML models and the associated hyperparameter search were implemented in Python using `scikit-learn`; XGBoost and Prophet were implemented using their dedicated libraries (`xgboost` and `prophet`, respectively), while the LSTM model was implemented using `Keras`.

V. EXPERIMENTAL EVALUATION

This section presents the empirical results obtained by applying the ML and DL models introduced in Section III-A. For each regressor, we explored a range of hyperparameter configurations and evaluated performance using the Mean Squared Error (MSE) metric. The models considered include LR, RF, XGBoost, Prophet, and LSTM.

A. Hyperparameter Tuning and Model Selection.

TABLE I: Hyperparameters explored for each model

Approach	Hyperparam	Range
LR	-	-
RF	estimators	100, 200, 500
	min_samples_leaf	None, 10, 20
	max_depth	5, 10, 15
XGBoost	estimators	100, 300, 500
	learning_rate	0.01, 0.1, 0.2
	max_depth	3, 9, 10, 11
	min_child_weight	1, 5
Prophet	-	-
LSTM	-	-

Hyperparameter optimization was conducted through a systematic grid-based search, evaluating multiple parameter combinations for each model on the EBITDA indicator. This approach enabled a thorough exploration of the hyperparameter space and supported an effective trade-off between predictive accuracy and model complexity. In particular, the search ranges were selected according to widely adopted best practices, so as to cover the complexity while keeping the combinatorial search computationally feasible. The ranges explored for each model are summarized in Table I.

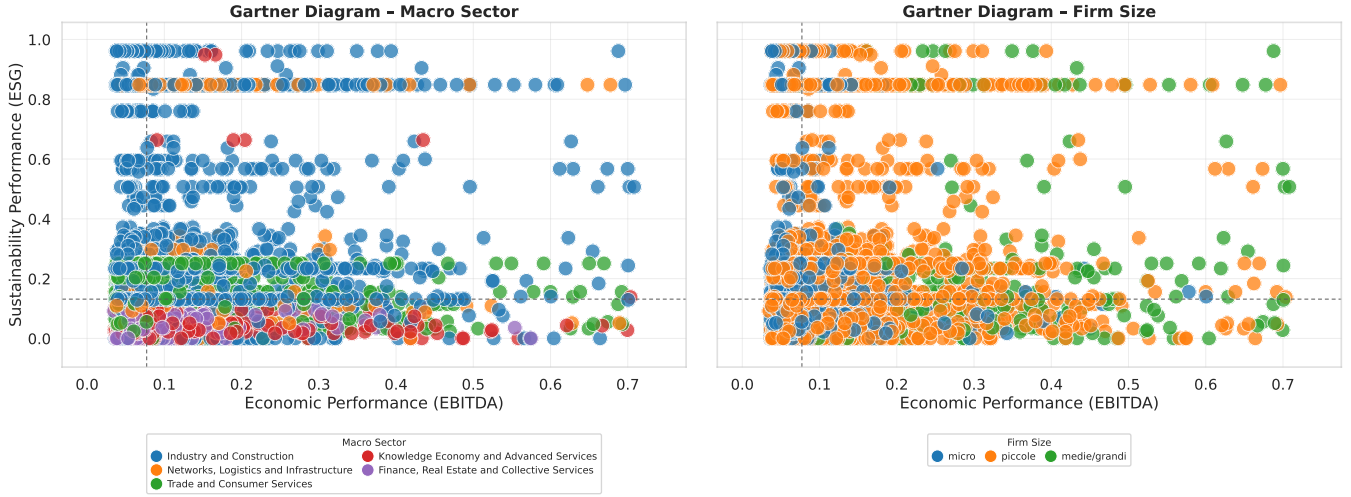


Fig. 2: Gartner-style diagram for firm positioning based on predicted EBITDA and ESG score.

TABLE II: MSE value for each model

Approach	MSE
LR	0.00299
RF	0.00285
XGBoost	0.00292
Prophet	0.00392
LSTM	0.00287

Selected Hyperparameter Settings. For each algorithm, we selected the optimal configuration as the one minimizing the MSE. In particular, the best-performing Random Forest model was obtained with $n_{\text{estimators}} = 100$, $max_depth = 10$, and $min_samples_leaf = 10$, while XGBoost achieved its best performance with $learning_rate = 0.01$, $max_depth = 10$, $n_{\text{estimators}} = 500$, and $min_child_weight = 1$. For Linear Regression, Prophet, and LSTM, no hyperparameter tuning was performed; instead, we adopted a fixed set of model parameters. This choice was motivated by the limited length of the available annual time series, which makes extensive hyperparameter search less stable for Prophet, and by the high dimensionality and computational cost of LSTM tuning, which may lead to unreliable gains. For Prophet, we fixed the model specification by disabling daily and weekly seasonalities and enabling yearly seasonality. For the LSTM model, we adopted the following predefined training setup: EPOCHS=32, TEST_SIZE=0.025, VAL_SIZE=0.20, and RANDOM_STATE=42.

Model Performance Comparison. The MSE scores for all evaluated models are reported in Table II.

When comparing the performance of the ML and DL models, Random Forest and LSTM achieved the lowest MSE, with nearly identical error values. XGBoost and Linear Regression also performed well, although with slightly higher errors. The competitive performance of Linear Regression may be partially explained by the preprocessing pipeline: missing values

were imputed via linear interpolation over time, which can induce approximately linear temporal patterns and therefore favor linear models. Finally, Prophet yielded the highest MSE on our dataset. This outcome is consistent with Prophet’s typical strengths and limitations: it performs best when the target exhibits relatively structured temporal dynamics (e.g., clear trend and seasonality), whereas predictive accuracy tends to degrade when the time series at the single-company level is short and/or does not exhibit strong seasonality.

Consequently, the subsequent analyses were conducted using the tuned Random Forest model, since, given comparable predictive performance, we opted for the less complex approach.

B. Random Forest Evaluation across Targets and Clusters.

After selecting Random Forest as the best-performing model, we employed it to perform regression on four target indicators: Sales Revenue, EBITDA, Net Income, and the ESG score.

Specifically, we aimed to predict the value of each indicator for the last available year (2024) using the historical observations from 2015 to 2023.

How Does the Error Change Based on Company Size, Geographic Area and Sector? To assess whether model performance was consistent across the categories used to group the data, we analyzed how the MSE varies by company size, geographic macro-area, and macro-sector. We observed that MSE values across geographic macro-areas are comparable, and a similar pattern holds across macro-sectors. With respect to company size, the MSE for micro firms is approximately one order of magnitude lower ($MSE = 0.00076$) than for small and medium-large firms (with $MSE = 0.00219$ and $MSE = 0.00505$, respectively). This indicates that the model performs best on micro companies and relatively worse on small and medium-large firms, although the MSE remains low in all three groups.

How Does Trend Classification Change Based on Employment Dynamics? To characterize firms’ growth trends, we adopt an employment-based classification approach grounded in official statistical definitions. Specifically, we analyze the evolution of the number of employees over a three-year horizon and compute the average annualized growth rate using a compound annual growth rate (CAGR) formulation. In line with the Eurostat definition of high-growth enterprises [10], firms exhibiting an average annual employment growth greater than 10% over the considered three-year period are classified as *in growth*, while firms with an average annual decline lower than −10% are classified as *in decline*. Firms whose employment growth rate falls within the interval [−10%, +10%] are labeled as *stable*, representing cases where variations are not considered economically material. The resulting classification outcomes, aggregated at the macro-sector level, are summarized in Table III, which reports the empirical distribution of firms across growth, stability, and decline categories. Notably, the prevalence of firms classified as growing is higher within the Trade and Consumer Services sector as well as in the Industry and Construction macro-sector, indicating comparatively stronger employment expansion dynamics in these areas.

TABLE III: Company classification by macro-sector and employment class

Macro-sector	declining	growing	stable
Trade and Consumer	180	1003	1569
Finance, Real Estate and Collective	26	133	283
Industry and Construction	241	1038	1831
Knowledge Economy and Advanced	47	268	295
Networks, Logistics and Infrastructure	59	266	421

How can we identify sustainable investment opportunities? Fig.2 illustrates a Gartner-style two-dimensional positioning of firms based on their predicted economic and sustainability performance. Specifically, firms are mapped onto the plane defined by predicted EBITDA, used as a proxy for future economic performance, and predicted ESG score, representing expected sustainability performance. Median-based thresholds along both dimensions partition the diagram into four areas, enabling a clear differentiation between firms exhibiting strong, weak, or asymmetric performance profiles. As shown in Fig.2, firms positioned closer to the region associated with simultaneously high predicted economic and ESG performance may represent promising opportunities for responsible and sustainable investment strategies. Notably, this region is predominantly populated by firms operating in the Industry and Construction macro-sector. Conversely, firms characterized by strong predicted economic performance but comparatively weaker ESG outcomes tend to occupy areas reflecting an imbalanced performance profile; such a configuration may expose them to long-term risks related to regulatory pressure, stakeholder expectations, or sustainability-related constraints. An illustrative example of this pattern is observed in the Commerce and Consumer Services sector.

VI. CONCLUSION

In this paper, we presented a comprehensive and interpretable framework for the predictive analysis of economic and sustainability performance in privately owned companies. By leveraging a rich company-level dataset and benchmarking multiple ML and DL models, we identified Random Forest as the most accurate approach for forecasting both financial indicators and ESG scores. The proposed methodology integrates robust preprocessing, forward-looking prediction, and model evaluation across heterogeneous company groups defined by size, macro-sector, and geographic aggregation.

Overall, our results highlight the value of combining machine learning, explainability, and structured visualization to address the complexity and heterogeneity of private company populations. Future research directions include extending the analysis to additional geographic contexts, incorporating alternative sustainability metrics, and exploring dynamic thresholding strategies to further refine company classification and investment-oriented insights.

ACKNOWLEDGMENT

The authors would like to thank Professor A. Montieri for the course “Tecnologie e Applicazioni per la Trasformazione Digitale” in which this project was developed and for providing the scientific guidance and passion that shaped our research. We are also grateful to Davide Di Monda, PhD student at the University of Naples Federico II, for the continuous support, valuable feedback, and practical advice provided throughout all phases of the project.

REFERENCES

- [1] M. Bianchini and V. Michalkova, “Data analytics in smes: Trends and policies,” 2019.
- [2] OECD, “Financing smes and entrepreneurs 2024: An oecd scoreboard,” 2024.
- [3] I. Guarino, A. Montieri, D. Ciunzo, and A. Pescapè, “Explainable predictive analysis of the economic and financial performance of italian publicly owned companies,” in *2025 International Conference on Artificial Intelligence in Information and Communication (ICAIIIC)*, 2025, pp. 0087–0092.
- [4] O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu, “Financial time series forecasting with deep learning : A systematic literature review: 2005–2019,” *Applied Soft Computing*, vol. 90, p. 106181, 2020.
- [5] I. Tarmuji, R. Maelah, and N. H. Tarmuji, “The impact of environmental, social and governance practices (esg) on economic performance: Evidence from esg score,” *International Journal of Trade, Economics and Finance*, vol. 7, no. 3, p. 67, 2016.
- [6] M.-A. Diaye, S.-H. Ho, and R. Oueghli, “Esg performance and economic growth: a panel co-integration analysis,” *Empirica*, vol. 49, no. 1, pp. 99–122, 2022.
- [7] V. D’Amato, R. D’Ecclesia, and S. Levantesi, “Esg score prediction through random forest algorithm,” *Computational Management Science*, vol. 19, no. 2, pp. 347–373, 2022.
- [8] M. A. F. Chowdhury, M. Abdullah, M. A. K. Azad, Z. Sulong, and M. N. Islam, “Environmental, social and governance (esg) rating prediction using machine learning approaches,” *Annals of Operations Research*, pp. 1–25, 2023.
- [9] S. J. Taylor and B. Letham, “Forecasting at scale,” *The American Statistician*, vol. 72, no. 1, pp. 37–45, 2018.
- [10] Eurostat, “High-growth enterprises – statistics,” Statistics Explained, European Commission, 2026.